

Differential Expression

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Contents

1. Import files	1
2. Differential Expression Analysis	2
3. Consolidate DEG lists	7
Session information	8

```
# Load necessary packages
library(DESeq2)
library(EnhancedVolcano)
library(tidyverse)
library(cowplot)
library(pheatmap)
library(viridis)
library(ggrepel)

# TIFF output directory (created automatically)
tiff_dir <- "../results/TIFFs/3_DifferentialExpression"
dir.create(tiff_dir, showWarnings = FALSE, recursive = TRUE)
```

1. Import files

We load the normalized dds object created during the preprocessing and batch correction steps.

```
load(file = "../results/dds_norm.RData")

dds_norm

## class: DESeqDataSet
## dim: 4566 4
## metadata(1): version
## assays(4): counts mu H cooks
## rownames(4566): lcl|NZ_JBCFOM010000001.1_cds_AAGW04_RS00515_104
##   lcl|NZ_JBCFOM010000001.1_cds_AAGW04_RS01695_335 ...
##   lcl|NZ_JBCFOM010000085.1_cds_WP_342414474.1_4391
##   lcl|NZ_JBCFOM010000086.1_cds_WP_342414475.1_4392
## rowData names(22): baseMean baseVar ... deviance maxCooks
## colnames(4): C1 C2 T1 T2
## colData names(4): Condition Replicates batch sizeFactor
```

We load the metadata that contains the information about the projects and experimental conditions.

```
metadatos <- read.csv(file = "../data/metadata_all.csv", row.names = 1)
head(metadatos)
```

```
##      Condition Replicates batch
## C1      Control      C1_R1      1
## C2      Control      C1_R2      2
## T1 Treatment      T1_R1      1
## T2 Treatment      T1_R2      2
```

We load the eggNOG functional annotation to map long gene IDs to readable short names (e.g., `pe1A`, `cysK`).

```
# Gene-name labels are optional: on a first pass this file does not exist yet and
# the volcano falls back to WP_ accessions; re-knit after script 4 to get gene symbols.
# Load eggNOG annotation if available (generated by script 4).
# If not yet present, label_map is empty and make_labels falls back to WP_ IDs.
annot_path <- "../results/advanced_analysis/full_annotated_results.csv"
if (file.exists(annot_path)) {
  full_annotation <- read_csv(annot_path, show_col_types = FALSE)
  label_map <- full_annotation %>%
    filter(!is.na(Preferred_name), !Preferred_name %in% c("NA", "-", "")) %>%
    dplyr::select(GeneID, Preferred_name) %>%
    distinct(GeneID, .keep_all = TRUE)
} else {
  message("Note: full_annotated_results.csv not found - run script 4 first for gene name labels.")
  label_map <- tibble(GeneID = character(), Preferred_name = character())
}

# Helper functions
short_wp <- function(x) str_extract(x, "WP_[0-9]+\\.?.[0-9]+")
make_labels <- function(long_ids) {
  wp <- short_wp(long_ids)
  labels <- ifelse(wp %in% label_map$GeneID,
    label_map$Preferred_name[match(wp, label_map$GeneID)],
    wp)
  labels[is.na(labels)] <- wp[is.na(labels)]
  labels
}

# Consistent condition color palette (used across all plots)
pal_control <- "#4575B4"
pal_treatment <- "#D73027"
cond_palette <- c("Control" = pal_control, "Treatment" = pal_treatment)
pal_diverging <- colorRampPalette(c("#4575B4", "white", "#D73027"))(100)
```

2. Differential Expression Analysis

We check the conditions available in our metadata to set up the comparisons.

```

metadatos %>%
  group_by(Condition) %>%
  summarise(Count = n())

```

```

## # A tibble: 2 x 2
##   Condition Count
##   <chr>      <int>
## 1 Control      2
## 2 Treatment    2

```

We create the DESeq2 results object, clearly defining the comparison: Treatment vs Control (positive log2FC = higher in Treatment). *Note: In DESeq2, the standard format is `c("factor", "numerator", "denominator")`.*

```

res <- results(dds_norm, contrast = c("Condition", "Treatment", "Control"))
res

```

```

## log2 fold change (MLE): Condition Treatment vs Control
## Wald test p-value: Condition Treatment vs Control
## DataFrame with 4566 rows and 6 columns
##
##                                     baseMean log2FoldChange
##                                     <numeric>   <numeric>
## lcl|NZ_JBCFOM010000001.1_cds_AAGW04_RS00515_104    288.801      0.354221
## lcl|NZ_JBCFOM010000001.1_cds_AAGW04_RS01695_335   3168.041     -0.491270
## lcl|NZ_JBCFOM010000001.1_cds_AAGW04_RS01865_369    1550.507      0.330091
## lcl|NZ_JBCFOM010000001.1_cds_WP_002208627.1_289     497.937      0.296170
## lcl|NZ_JBCFOM010000001.1_cds_WP_005971891.1_35   11678.569     -1.133319
## ...
## lcl|NZ_JBCFOM010000081.1_cds_AAGW04_RS23200_4388   4496.053      0.41002390
## lcl|NZ_JBCFOM010000082.1_cds_WP_342414473.1_4389   3390.522      0.10906346
## lcl|NZ_JBCFOM010000084.1_cds_AAGW04_RS23210_4390   3856.171     -0.01458072
## lcl|NZ_JBCFOM010000085.1_cds_WP_342414474.1_4391   4680.954     -0.09858314
## lcl|NZ_JBCFOM010000086.1_cds_WP_342414475.1_4392    309.714     -0.00512489
##
##                                     lfcSE      stat
##                                     <numeric>   <numeric>
## lcl|NZ_JBCFOM010000001.1_cds_AAGW04_RS00515_104    0.1886450      1.87771
## lcl|NZ_JBCFOM010000001.1_cds_AAGW04_RS01695_335    0.0576660     -8.51924
## lcl|NZ_JBCFOM010000001.1_cds_AAGW04_RS01865_369    0.0816777      4.04139
## lcl|NZ_JBCFOM010000001.1_cds_WP_002208627.1_289    0.1437417      2.06043
## lcl|NZ_JBCFOM010000001.1_cds_WP_005971891.1_35     0.0315276    -35.94691
## ...
## lcl|NZ_JBCFOM010000081.1_cds_AAGW04_RS23200_4388    0.0483513     8.4801093
## lcl|NZ_JBCFOM010000082.1_cds_WP_342414473.1_4389    0.0553935     1.9688871
## lcl|NZ_JBCFOM010000084.1_cds_AAGW04_RS23210_4390    0.0614663    -0.2372150
## lcl|NZ_JBCFOM010000085.1_cds_WP_342414474.1_4391    0.0482871    -2.0416046
## lcl|NZ_JBCFOM010000086.1_cds_WP_342414475.1_4392    0.1821094    -0.0281418
##
##                                     pvalue      padj
##                                     <numeric>   <numeric>
## lcl|NZ_JBCFOM010000001.1_cds_AAGW04_RS00515_104    6.04208e-02   9.69366e-02
## lcl|NZ_JBCFOM010000001.1_cds_AAGW04_RS01695_335    1.60604e-17   7.45240e-17
## lcl|NZ_JBCFOM010000001.1_cds_AAGW04_RS01865_369    5.31355e-05   1.26166e-04
## lcl|NZ_JBCFOM010000001.1_cds_WP_002208627.1_289    3.93574e-02   6.55621e-02
## lcl|NZ_JBCFOM010000001.1_cds_WP_005971891.1_35     5.65692e-283  1.83188e-281

```

```
## ...
## lcl|NZ_JBCFOM010000081.1_cds_AAGW04_RS23200_4388 2.24982e-17 1.03765e-16
## lcl|NZ_JBCFOM010000082.1_cds_WP_342414473.1_4389 4.89661e-02 8.00795e-02
## lcl|NZ_JBCFOM010000084.1_cds_AAGW04_RS23210_4390 8.12490e-01 8.80776e-01
## lcl|NZ_JBCFOM010000085.1_cds_WP_342414474.1_4391 4.11908e-02 6.83171e-02
## lcl|NZ_JBCFOM010000086.1_cds_WP_342414475.1_4392 9.77549e-01 9.85318e-01
```

```
# Export the full statistical results file
write.csv(res, file = "../results/deseq_results_Treatment_vs_Control.csv")
```

2.1. Volcano Plot

We create a Volcano plot to get a general view of the differential expression. We set the statistical significance limit ($\text{padj} < 0.01$) and the fold change limit ($|\log_2\text{FC}| > 1$).

```
# Map long row IDs (lcl|NZ_...) to short gene names (pelA, cysK, ...)
row_labels <- make_labels(rownames(res))

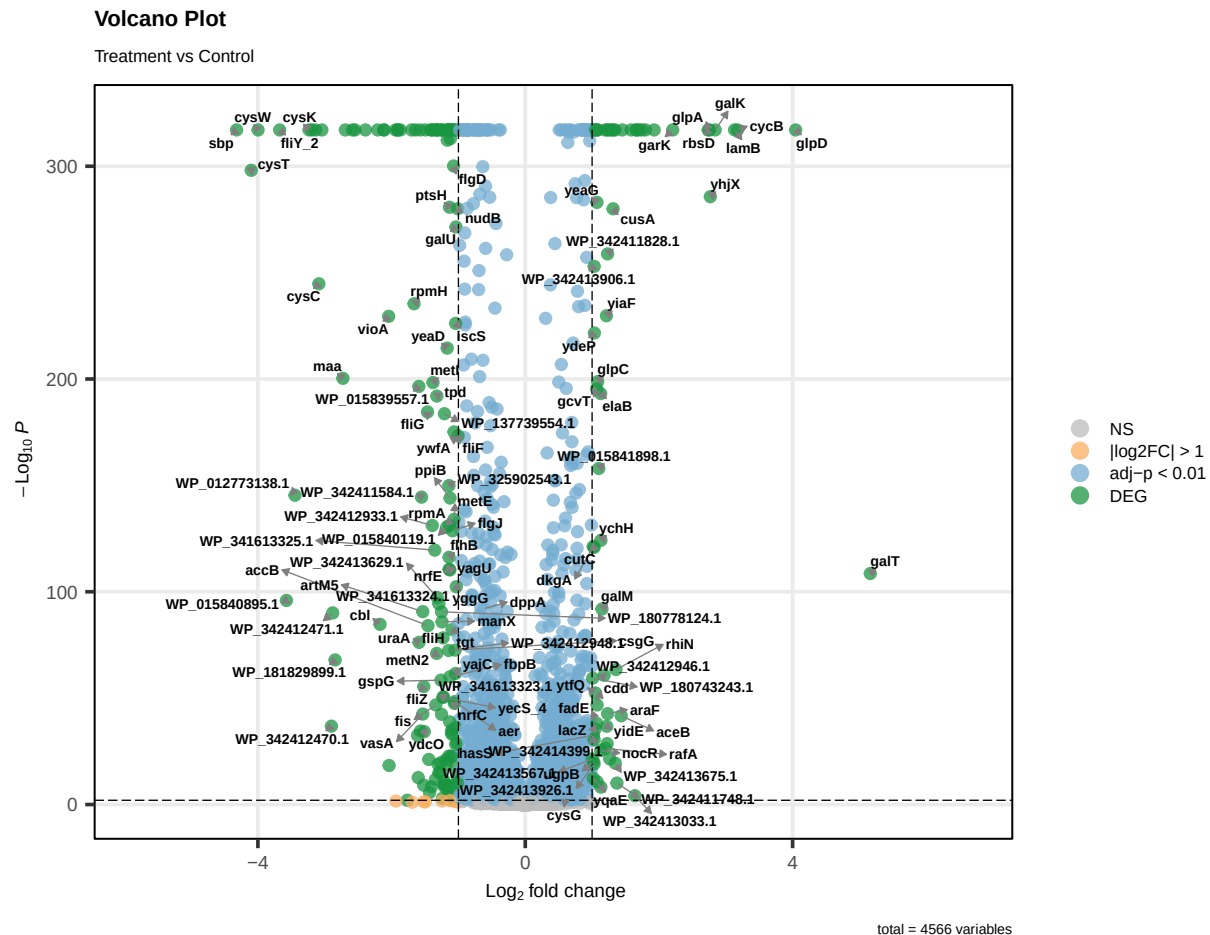
# Only annotate genes that pass both significance cutoffs
sig_mask <- !is.na(res$padj) & res$padj < 0.01 & abs(res$log2FoldChange) > 1
top_labels <- row_labels[sig_mask]

volc <- EnhancedVolcano(res,
  lab          = row_labels,      # short gene names
  selectLab    = top_labels,      # only label DEGs
  x            = "log2FoldChange",
  y            = "padj",
  axisLabSize  = 12,
  pCutoff      = 0.01,
  FCcutoff     = 1,
  col          = c("#BDBDBD",     # NS
                   "#FDAE61",     # |FC| only
                   "#74ADD1",     # p-val only
                   "#1A9850"),    # DEG (both)
  colAlpha     = 0.75,
  pointSize    = 2.5,
  title        = "Volcano Plot",
  titleLabSize = 14,
  subtitle     = "Treatment vs Control",
  subtitleLabSize = 11,
  captionLabSize = 9,
  labSize      = 3.2,
  labFace      = "bold",
  labCol       = "black",
  legendLabels = c("NS", "|log2FC| > 1", "adj-p < 0.01", "DEG"),
  legendPosition = "right",
  legendLabSize = 11,
  legendIconSize = 4,
  drawConnectors = TRUE,
  widthConnectors = 0.4,
  colConnectors = "grey50",
  border        = "full",
  borderWidth   = 0.6,
```

```

gridlines.major = TRUE,
gridlines.minor = FALSE)
volc

```



```

ggsave(filename = "../results/volcano_Treatment_vs_Control.pdf",
        plot = volc, width = 11, height = 8.5)
ggsave(filename = file.path(tiff_dir, "volcano_Treatment_vs_Control.tiff"),
        plot = volc, width = 11, height = 8.5, dpi = 300, compression = "lzw")

```

2.2. Extract Differentially Expressed Genes (DEGs)

We filter the results to only keep the genes that pass our significance and fold change limits.

```

# Strict filtering
deg <- subset(res, padj < 0.01 & abs(log2FoldChange) > 1)
paste("Total DEGs found:", nrow(deg))

```

```
## [1] "Total DEGs found: 216"
```

```
# Subset: Up-regulated genes in the 'Treatment' condition relative to 'Control'
up <- subset(deg, log2FoldChange > 1)
paste("Up-regulated genes:", nrow(up))
```

```
## [1] "Up-regulated genes: 68"
```

```
# Subset: Down-regulated genes
down <- subset(deg, log2FoldChange < (-1))
paste("Down-regulated genes:", nrow(down))
```

```
## [1] "Down-regulated genes: 148"
```

```
# Export DEG list
write.csv(deg, file = "../results/deg_Treatment_vs_Control.csv", row.names = TRUE)
```

2.3. Expression Heatmap

We generate a heatmap to see the expression patterns of the found DEGs across the samples.

```
# Extract normalized counts from dds_norm (assuming it was saved in the RData file)
normcounts <- counts(dds_norm, normalized = TRUE)
```

```
# Match column names with the replicates in the metadata
colnames(normcounts) <- metadatos$Replicates
```

```
# Dynamically select the samples that belong to this comparison
muestras_contraste <- metadatos %>%
  filter(Condition %in% c("Control", "Treatment")) %>%
  pull(Replicates)
```

```
subnormcounts <- normcounts[, muestras_contraste]
```

```
# Extract DEG IDs to filter the matrix
deg_id <- rownames(deg)
```

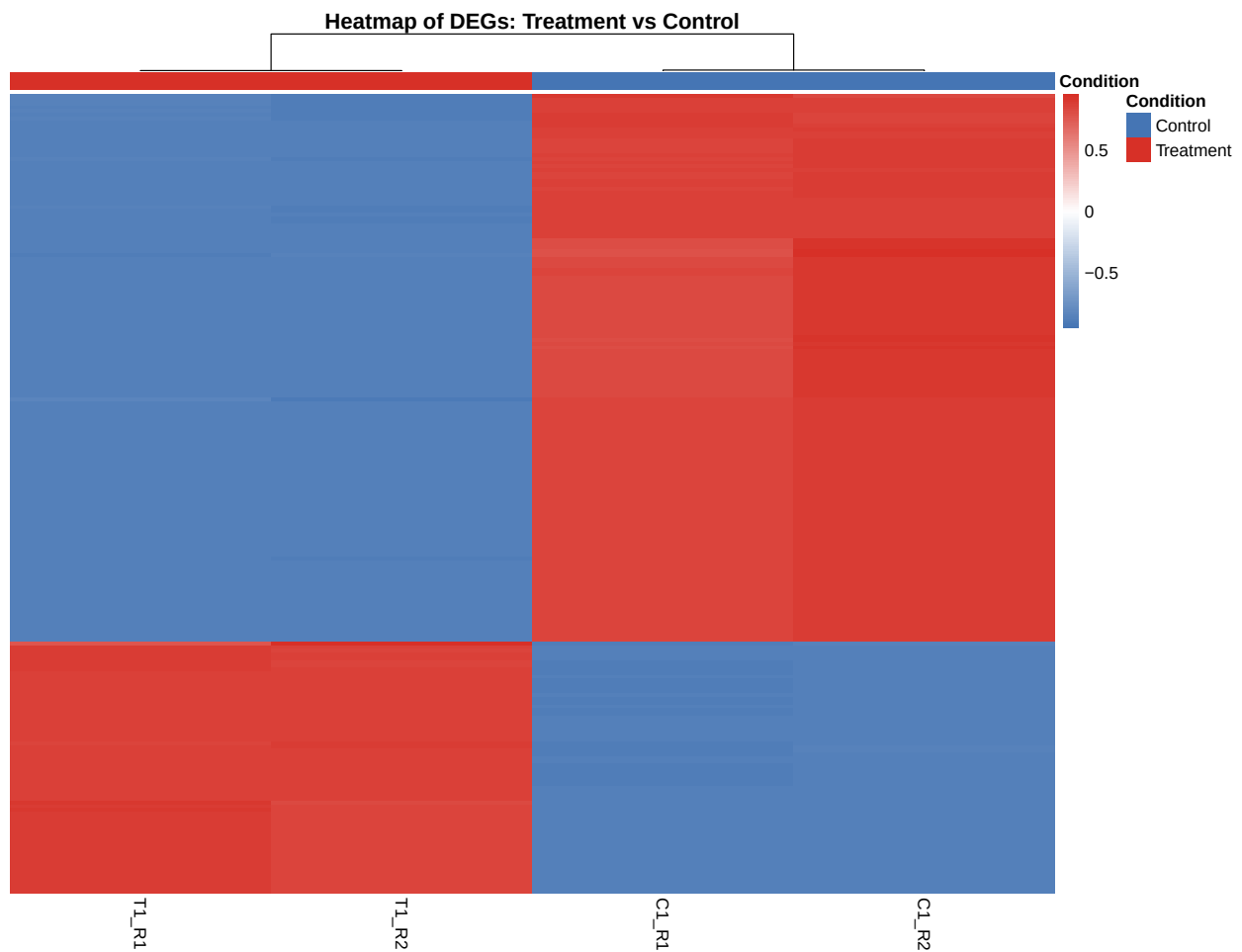
```
# Condition annotation bar for columns
col_annot <- data.frame(Condition = metadatos$Condition,
  row.names = metadatos$Replicates)
annot_colors <- list(Condition = cond_palette)
```

```
# Create the heatmap with row scaling (Z-score) and diverging Blue+White+Red palette
hm <- pheatmap(subnormcounts[deg_id, ],
  scale = "row",
  color = pal_diverging,
  show_rownames = FALSE,
  treeheight_row = 0,
  treeheight_col = 20,
  annotation_col = col_annot,
  annotation_colors = annot_colors,
  main = "Heatmap of DEGs: Treatment vs Control",
  fontsize = 11,
```

```

border_color = NA,
silent       = TRUE)
hm

```



```

ggsave(filename = "../results/heatmap_Treatment_vs_Control.pdf",
        plot = hm, width = 11, height = 8.5)
ggsave(filename = file.path(tiff_dir, "heatmap_Treatment_vs_Control.tiff"),
        plot = hm, width = 11, height = 8.5, dpi = 300, compression = "lzw")

```

3. Consolidate DEG lists

Since we are analyzing a single main comparison in this script, we save the DEG IDs in a clean table format for future functional analysis.

(Note: This section is ready to scale up if you add lists from other projects later).

```

# Load the generated list
BP1_deg <- read.csv(file = "../results/deg_Treatment_vs_Control.csv", row.names = 1)

```

```
# Create a data frame with the IDs
deg_summary <- data.frame(
  GeneID = rownames(BP1_deg),
  Contrast = "Treatment_vs_Control"
)
```

```
head(deg_summary)
```

```
##                               GeneID          Contrast
## 1 1c1|NZ_JBCFOM010000001.1_cds_WP_005971891.1_35 Treatment_vs_Control
## 2 1c1|NZ_JBCFOM010000001.1_cds_WP_012773630.1_175 Treatment_vs_Control
## 3 1c1|NZ_JBCFOM010000001.1_cds_WP_012773703.1_250 Treatment_vs_Control
## 4 1c1|NZ_JBCFOM010000001.1_cds_WP_015839418.1_354 Treatment_vs_Control
## 5 1c1|NZ_JBCFOM010000001.1_cds_WP_015839489.1_420 Treatment_vs_Control
## 6 1c1|NZ_JBCFOM010000001.1_cds_WP_039348726.1_36  Treatment_vs_Control
```

```
# Export the table ready for enrichment or network analysis
write_csv(deg_summary, file = "../results/common_unique_degs.csv")
```

Session information

```
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=Spanish_Mexico.utf8  LC_CTYPE=Spanish_Mexico.utf8
## [3] LC_MONETARY=Spanish_Mexico.utf8 LC_NUMERIC=C
## [5] LC_TIME=Spanish_Mexico.utf8
##
## time zone: America/Mexico_City
## tzcode source: internal
##
## attached base packages:
## [1] stats4      stats      graphics  grDevices  utils      datasets  methods
## [8] base
##
## other attached packages:
## [1] pdftools_3.9.0          tidygraph_1.3.1
## [3] ggraph_2.2.2           igraph_2.2.2
## [5] KEGGREST_1.50.0        magick_2.9.1
## [7] STRINGdb_2.22.0        UpSetR_1.4.1
## [9] GO.db_3.22.0            AnnotationDbi_1.72.0
```

```

## [11] pathview_1.50.0          enrichplot_1.30.5
## [13] clusterProfiler_4.18.4   viridis_0.6.5
## [15] viridisLite_0.4.3        pheatmap_1.0.13
## [17] cowplot_1.2.0            lubridate_1.9.5
## [19] forcats_1.0.1            stringr_1.6.0
## [21] dplyr_1.2.0              purrr_1.2.1
## [23] readr_2.2.0              tidyr_1.3.2
## [25] tibble_3.3.1             tidyverse_2.0.0
## [27] EnhancedVolcano_1.28.2   ggrepel_0.9.7
## [29] ggplot2_4.0.3            DESeq2_1.50.2
## [31] SummarizedExperiment_1.40.0 Biobase_2.70.0
## [33] MatrixGenerics_1.22.0    matrixStats_1.5.0
## [35] GenomicRanges_1.62.1     Seqinfo_1.0.0
## [37] IRanges_2.44.0           S4Vectors_0.48.0
## [39] BiocGenerics_0.56.0      generics_0.1.4
##
## loaded via a namespace (and not attached):
## [1] splines_4.5.2             bitops_1.0-9              ggplotify_0.1.3
## [4] R.oo_1.27.1               polyclip_1.10-7           graph_1.88.1
## [7] XML_3.99-0.23             lifecycle_1.0.5           lattice_0.22-7
## [10] vroom_1.7.1              MASS_7.3-65              magrittr_2.0.4
## [13] rmarkdown_2.31           plotrix_3.8-14            yaml_2.3.12
## [16] otl_0.2.0                 ggtangle_0.1.2            askpass_1.2.1
## [19] DBI_1.3.0                 RColorBrewer_1.1-3        abind_1.4-8
## [22] R.utils_2.13.0           RCurl_1.98-1.19          hash_2.2.6.4
## [25] yulab.utils_0.2.4        tweenr_2.0.3              rappdirs_0.3.4
## [28] gdtools_0.5.0            tidytree_0.4.8           proto_1.0.0
## [31] codetools_0.2-20         DelayedArray_0.36.1       DOSE_4.4.0
## [34] ggforce_0.5.0            tidyselect_1.2.1         aplot_0.3.1
## [37] farver_2.1.2             jsonlite_2.0.0            ggribes_0.5.7
## [40] systemfonts_1.3.2        chron_2.3-63              tools_4.5.2
## [43] ggnewscale_0.5.2         treeio_1.34.0            ragg_1.5.0
## [46] snow_0.4-4               Rcpp_1.1.1               glue_1.8.0
## [49] gridExtra_2.3.1          SparseArray_1.10.10       xfun_0.55
## [52] qvalue_2.42.0            withr_3.0.3              fastmap_1.2.0
## [55] caTools_1.18.4           digest_0.6.39            timechange_0.4.0
## [58] R6_2.6.1                 gridGraphics_0.5-1       qpdf_1.4.1
## [61] textshaping_1.0.4        gtools_3.9.5             RSQLite_2.4.6
## [64] R.methodsS3_1.8.2        utf8_1.2.6               fontLiberation_0.1.0
## [67] data.table_1.18.2.1      graphlayouts_1.2.5       httr_1.4.8
## [70] htmlwidgets_1.6.4        S4Arrays_1.10.1          scatterpie_0.2.6
## [73] sqldf_0.4-12            pkgconfig_2.0.3          gtable_0.3.6
## [76] blob_1.3.0              S7_0.2.1                 XVector_0.50.0
## [79] htmltools_0.5.9         fontBitstreamVera_0.1.1 fgsea_1.36.2
## [82] scales_1.4.0            png_0.1-9                ggfun_0.2.1
## [85] knitr_1.51              tzdb_0.5.0               reshape2_1.4.5
## [88] curl_7.0.0              nlme_3.1-168             org.Hs.eg.db_3.22.0
## [91] cachem_1.1.0            KernSmooth_2.23-26       parallel_4.5.2
## [94] pillar_1.11.1           grid_4.5.2               vctrs_0.7.1
## [97] gplots_3.3.0            tidydr_0.0.6             cluster_2.1.8.1
## [100] Rgraphviz_2.54.0         evaluate_1.0.5           KEGGgraph_1.70.0
## [103] gsubfn_0.7              tinytex_0.60             cli_3.6.5
## [106] locfit_1.5-9.12         compiler_4.5.2           rlang_1.3.0
## [109] crayon_1.5.3            labeling_0.4.3           plyr_1.8.9

```

## [112] fs_2.1.0	ggiraph_0.9.6	stringi_1.8.7
## [115] BiocParallel_1.44.0	Biostrings_2.78.0	lazyeval_0.2.2
## [118] GOSemSim_2.36.0	fontquiver_0.2.1	Matrix_1.7-4
## [121] hms_1.1.4	patchwork_1.3.2	bit64_4.6.0-1
## [124] memoise_2.0.1	ggtree_4.0.5	fastmatch_1.1-8
## [127] bit_4.6.0	ape_5.8-1	gson_0.2.0